REGRESSION ANALYSIS · UNIT 2

Logistic Regression

Binary, nominal, and ordinal variables

Data Science Engineering | Prof. María Elisa Vaca

MOTIVATION

Two different questions, two different models

LINEAR REGRESSION

How much is Y worth?

- **Continuous response variable Y** (e.g., credit line amount)
- Fitted by minimizing the sum of **squared errors** (least squares)
- Evaluated with **R^2 and significance tests** (p-values) on the coefficients
- The **line is used to predict** the expected value of Y

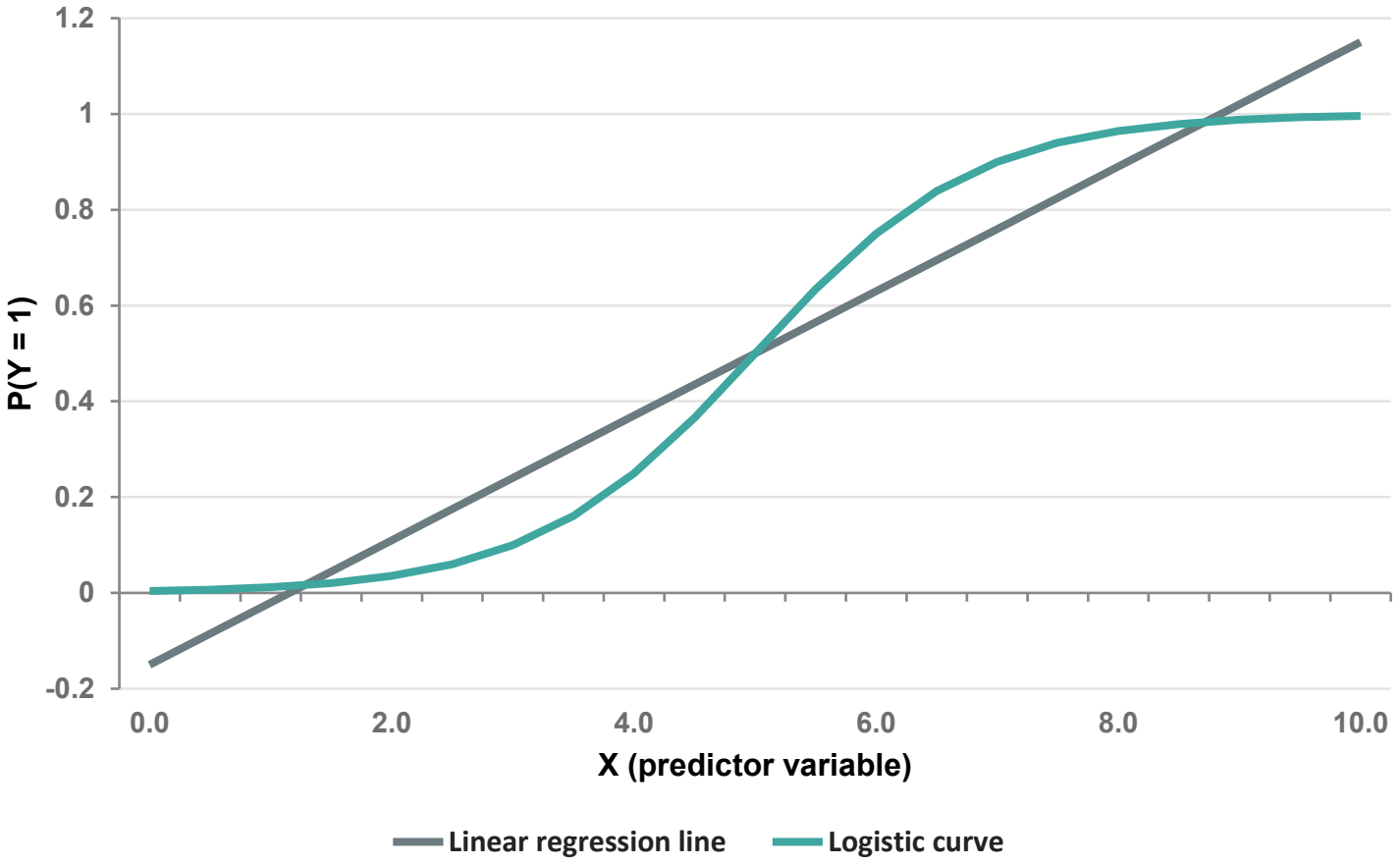
LOGISTIC REGRESSION

Does Y belong to group 1?

- **Binary response variable Y** (e.g., did the client pay late? yes/no)
- Fitted by **maximizing the likelihood** (maximum likelihood)
- Evaluated with **pseudo-R^2, a confusion matrix, and z-tests** on the coefficients
- The **curve is used to estimate the probability** of belonging to class 1

Both share the same spirit: finding the coefficients β that best explain Y from X.

The problem with fitting a straight line to binary data



- **The linear prediction can fall outside [0,1]:** numbers that are not valid probabilities.
- **Var(Y) = p(1-p) depends on p itself** the error variance is never constant (heteroscedasticity is guaranteed).
- **The real relationship between X and P(Y=1) is S-shaped**, not a straight line.
- We need a function that "squashes" any real value into the interval (0,1).

INTUITION

What are odds?

- Imagine your team has **1-to-4 odds**: out of every 5 similar games, it wins 1 and loses 4.

PROBABILITY

What fraction of ALL games does it win?

$$1 / 5 = 0.20$$

wins : (wins + losses)

ODDS

How many times does it win FOR EVERY loss?

ODDS ≠ PROBABILITY

$$\text{odds} = 1 / 4 = 0.25$$

- Probability: compares the event against EVERYTHING that can happen (win + lose).
- Odds: compare the event ONLY against its opposite (win vs. lose, not against the total).
- In this example: probability = 0.20, but odds = 0.25, similar numbers but conceptually different.

INTUITION

Odds in Everyday Contexts

odds = successes : failures (not successes : total, like probability)

ODDS > 1 · IN FAVOR

Basketball free throws

8 makes : 2 misses

$$\text{odds} = 8 / 2 = 4.0$$

Makes happen 4× as often as misses.

ODDS = 1 · EVEN

Fair coin flip

1 heads : 1 tails

$$\text{odds} = 1 / 1 = 1.0$$

Equal chance for and against — the neutral point.

ODDS > 1 · IN FAVOR

Exam pass rate

18 pass : 2 fail (of 20)

$$\text{odds} = 18 / 2 = 9.0$$

Passing is 9× more likely than failing.

ODDS < 1 · AGAINST

Client loan default

15 default : 85 on-time (of 100)

$$\text{odds} = 15 / 85 \approx 0.18$$

Defaulting is far less likely than paying on time.

ODDS < 1 · AGAINST

Weekend rain forecast

3 rainy : 7 dry (of 10 days)

$$\text{odds} = 3 / 7 \approx 0.43$$

Rain is less likely than a dry day, but not rare.

ODDS < 1 · AGAINST

Rare disease screening

2 sick : 998 healthy (of 1,000)

$$\text{odds} = 2 / 998 \approx 0.002$$

Being sick is extremely unlikely in this group.

INTUITION

From probability to odds

- Now the odds are 5 to 3: out of 8 games, it wins 5 and loses 3.

- $\mathrm{P}(win) = \frac{5}{8} = 0.625$
- $\mathrm{P}(lose) = \frac{3}{8} = 0.375$

The odds are the ratio between the two:

$$\text{odds} = \frac{\frac{5}{8}}{\frac{3}{8}} = \frac{5}{3} = 1.7$$

For every loss, it wins 1.7 times.

GENERAL FORMULA

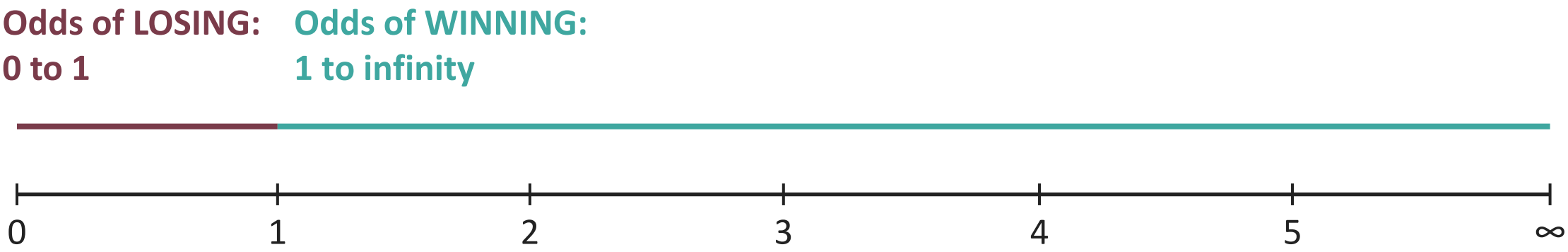
$$\frac{p}{1-p}$$

where p is the probability that the event occurs

- If p = 0.5 → odds = 1 (equal chance for and against)
- If p > 0.5 → odds > 1 (more in favor)
- If p < 0.5 → odds < 1 (more against)

Odds are asymmetric

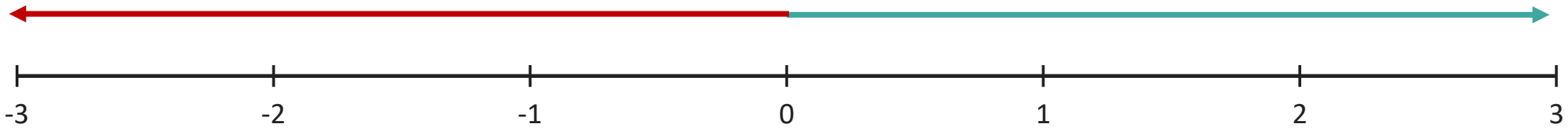
- Odds against range from 0 to 1.
- Odds in favor range from 1 to infinity. The scale is not symmetric around "even".



This asymmetry makes it hard to compare "how much in favor" against "how much against" something is and complicates using odds as a variable in a statistical model.

Log(Odds): a symmetric scale

Same distance in both directions



If the odds are 1 to 6 against:

$$\log\left(\frac{1}{6}\right) = -1.79$$

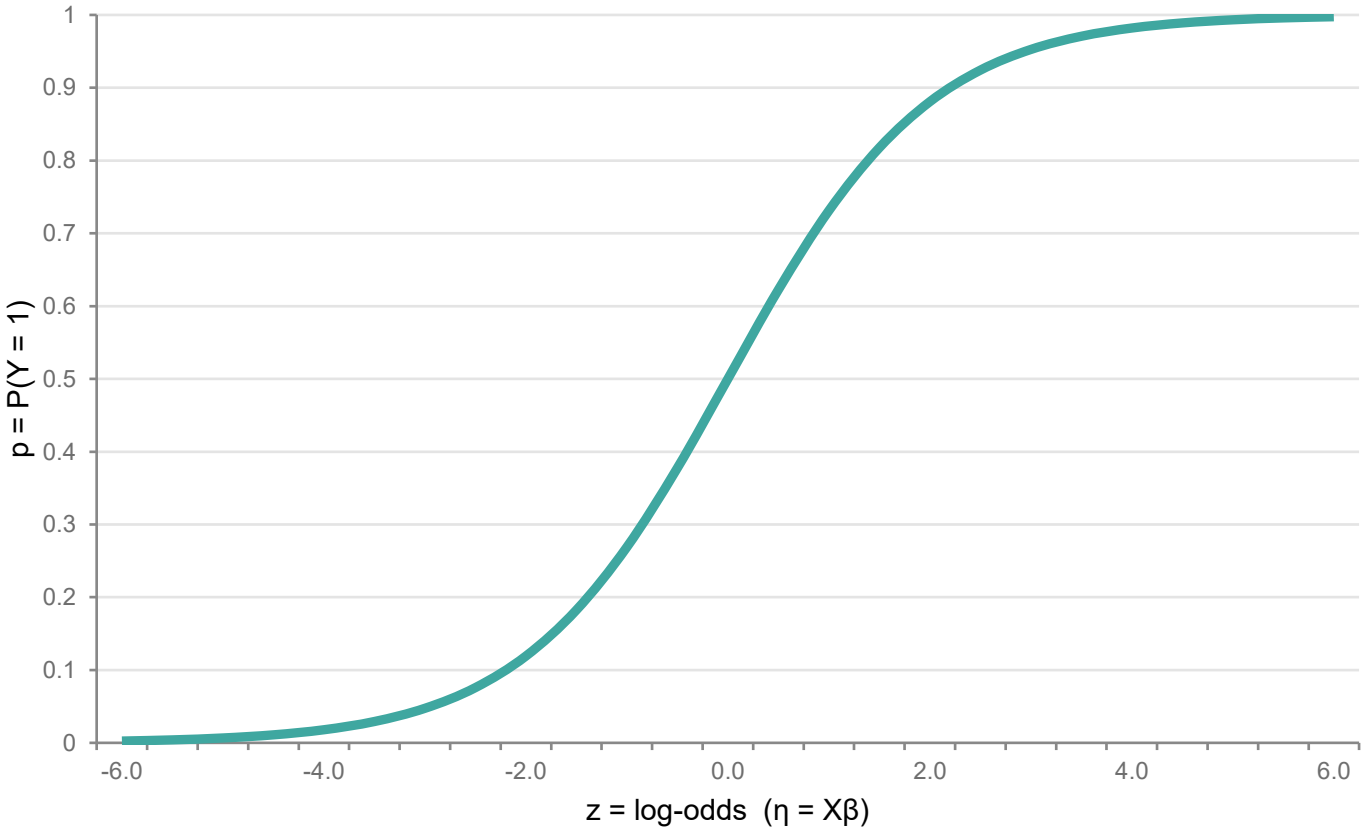
If the odds are 6 to 1 in favor:

$$\log\left(\frac{6}{1}\right) = +1.79$$

$$\text{logit}(p) = \ln\left(\frac{p}{1-p}\right)$$

- This transformation is called the logit function. It converts any probability into a real number symmetric around 0.
- The foundation of logistic regression.

The logistic function (sigmoid)



$$p = \frac{1}{1 + e^{-z}}$$

- No matter how negative or positive z is, the output always falls between 0 and 1.
- z = 0 → p = 0.5 (the curve's inflection point)
- Large positive z → p → 1
- Large negative z → p → 0
- This S-shape is exactly the inverse of the logit function we saw earlier.

Two paths, one relationship

LINEAR PREDICTOR

$$z = \beta_0 + \beta_1 x$$

lives in $(-\infty, +\infty)$

logistic function (sigmoid) →

← logit function (inverse)

PROBABILITY

$$p = P(Y = 1)$$

lives in (0, 1)

The link function is what connects both worlds

It transforms an unrestricted linear predictor into a valid probability, and vice versa.

In logistic regression, that **link function is the logit**, which is why it's also called the logit model.

GENERAL FRAMEWORK

Logistic regression as a Generalized Linear Model (GLM)

1

Random component

$$Y_i \sim \text{Bernoulli}(p_i)$$

How the response variable is distributed.

2

Systematic component

$$\eta_i = x_i^T \beta$$

The linear predictor: a combination of the X's.

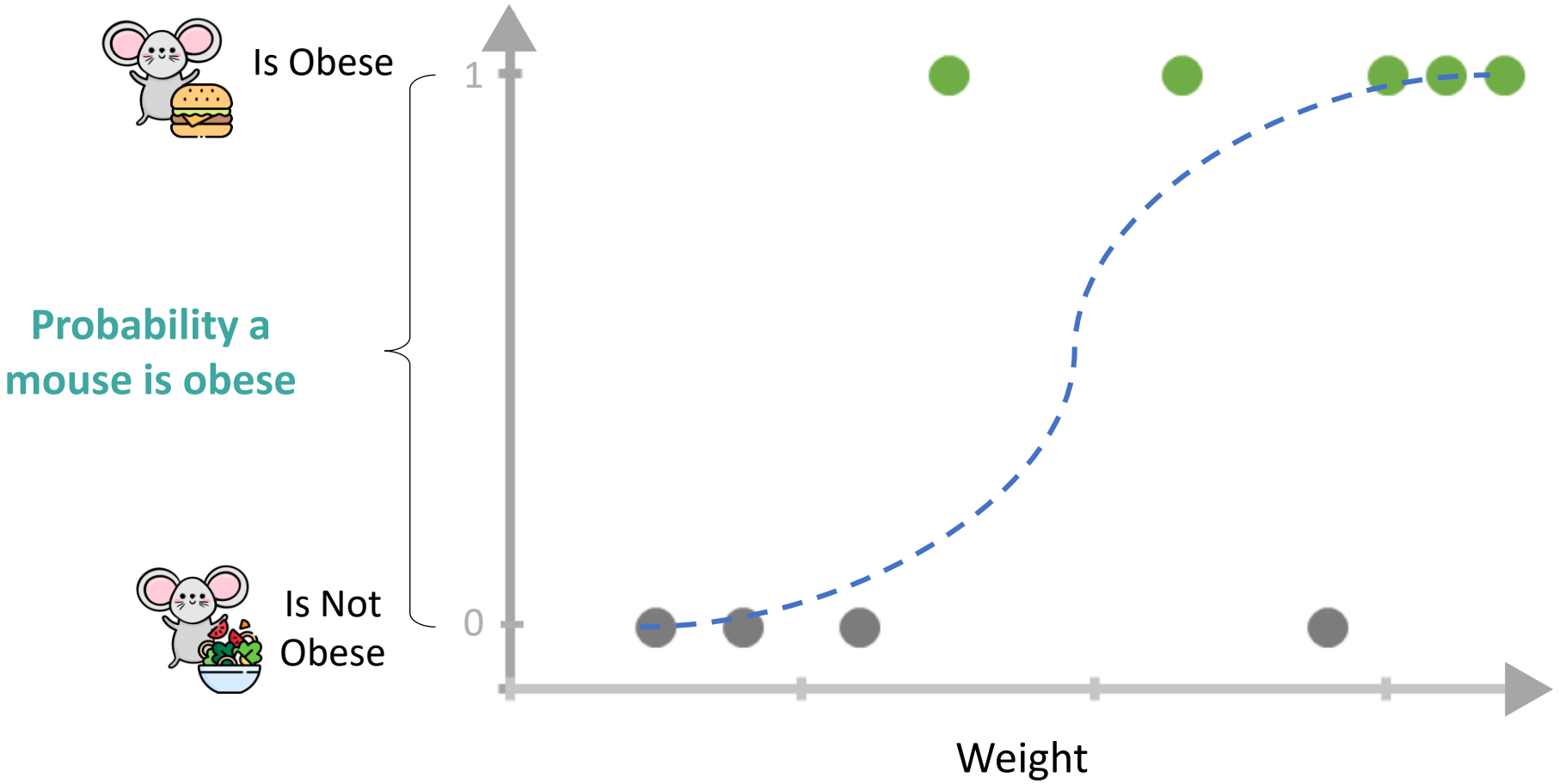
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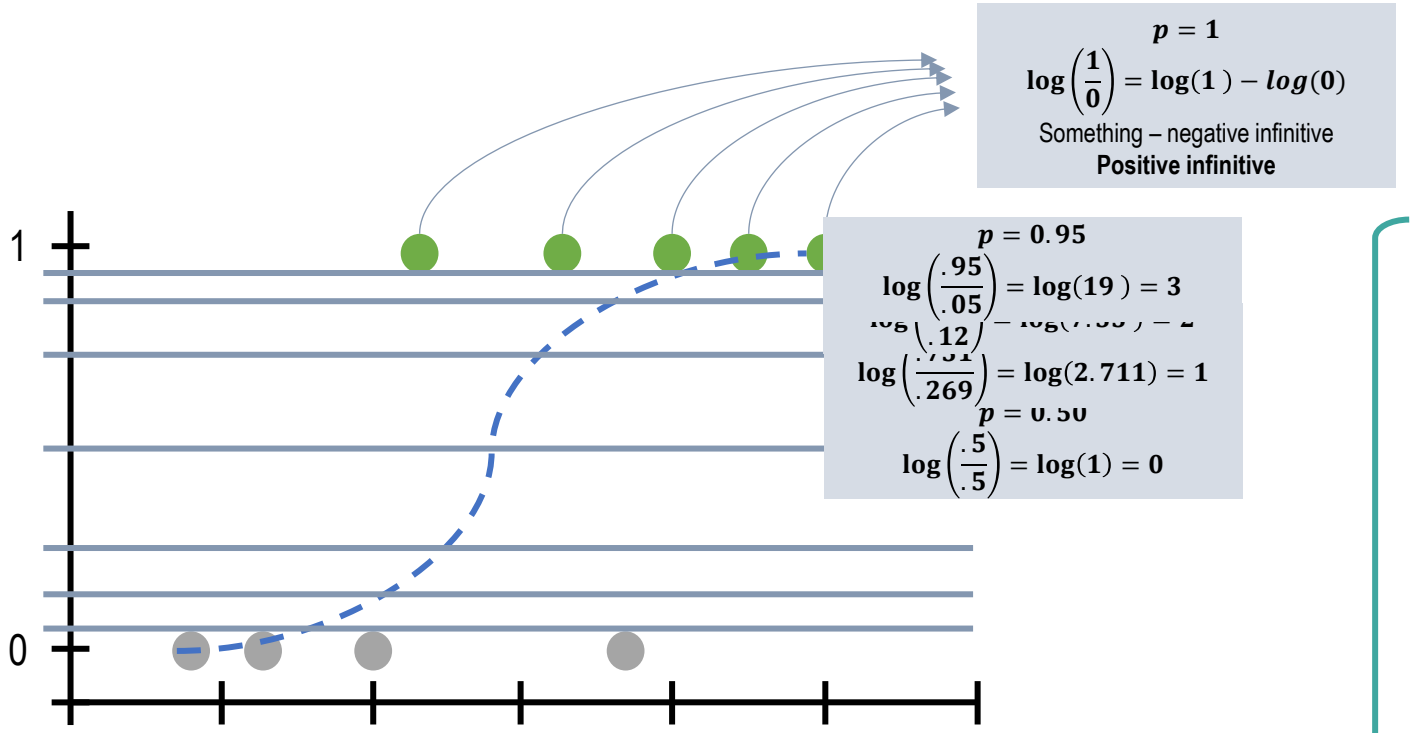
Link function

$$\text{logit}(p_i) = \eta_i$$

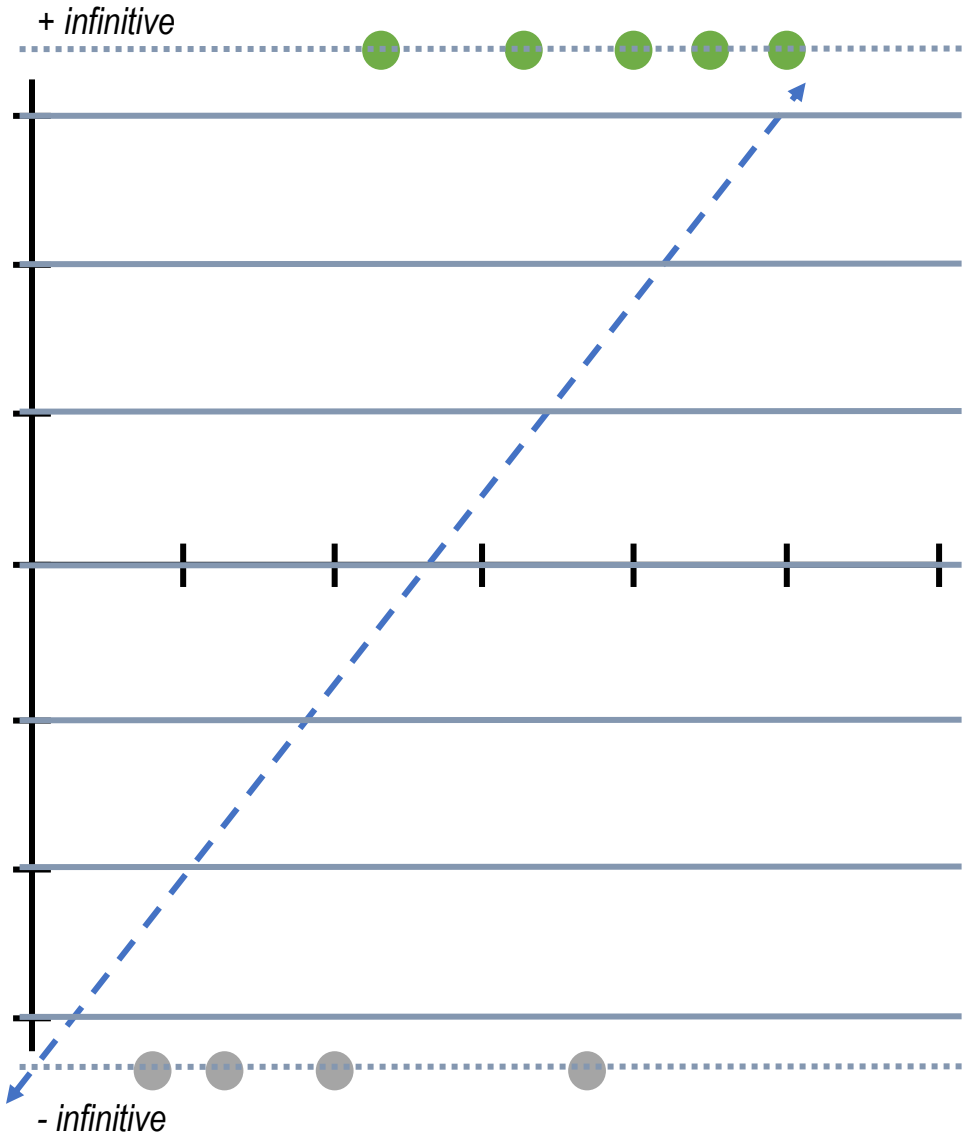
Connects the mean of Y with the linear predictor.

MICE EXAMPLE – CONTNUOUS VARIABLE





The coefficients are presented in terms of the **Log (Odds)**



Logit function

$$\log\left(\frac{p}{1-p}\right)$$

Example: weight and obesity in mice

	Estimate	Std. Error	z value	Pr(>|z|)
(Intercept)	-3.476	2.364	-1.471	0.1414
weight	1.825	1.088	1.678	0.0934

$$\hat{y} = -3.48 + 1.83 \cdot \text{weight}$$

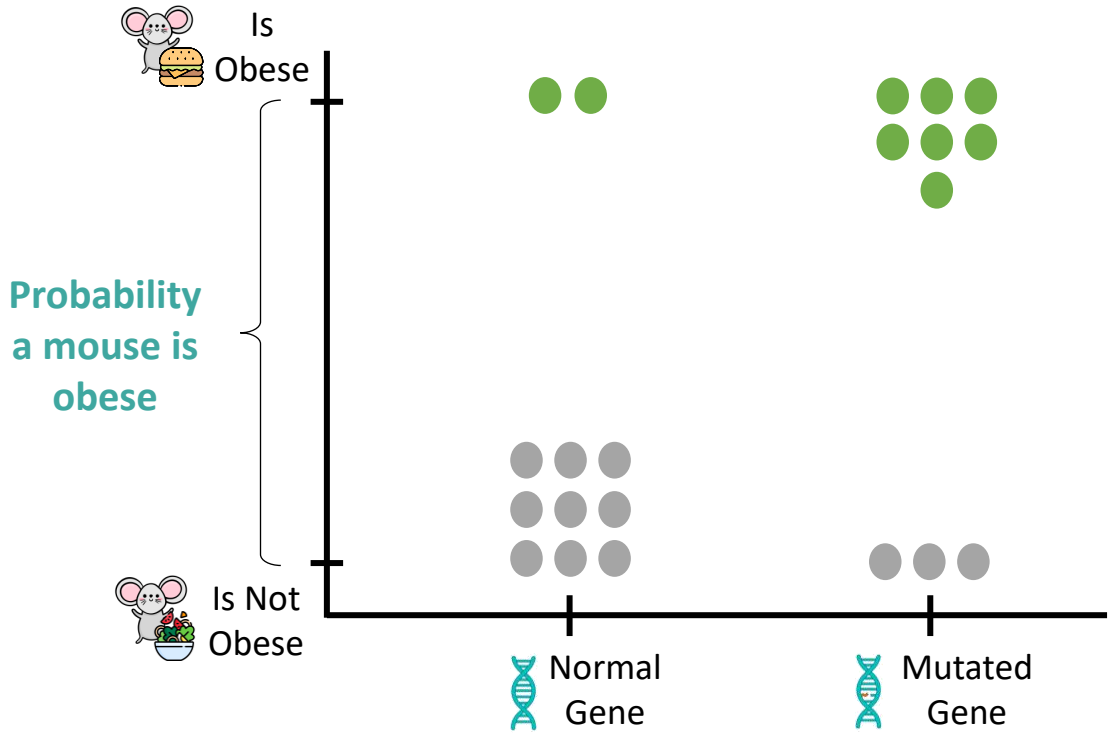
- **Intercept (–3.48):** the log-odds of being obese when weight = 0 (in this case, an extrapolation beyond the data range).
- Slope (1.83): **for each additional unit of weight**, the **log-odds of obesity increase by 1.83.**
- The positive sign already tells us the direction: **higher weight, higher probability of obesity.**

QUICK READ

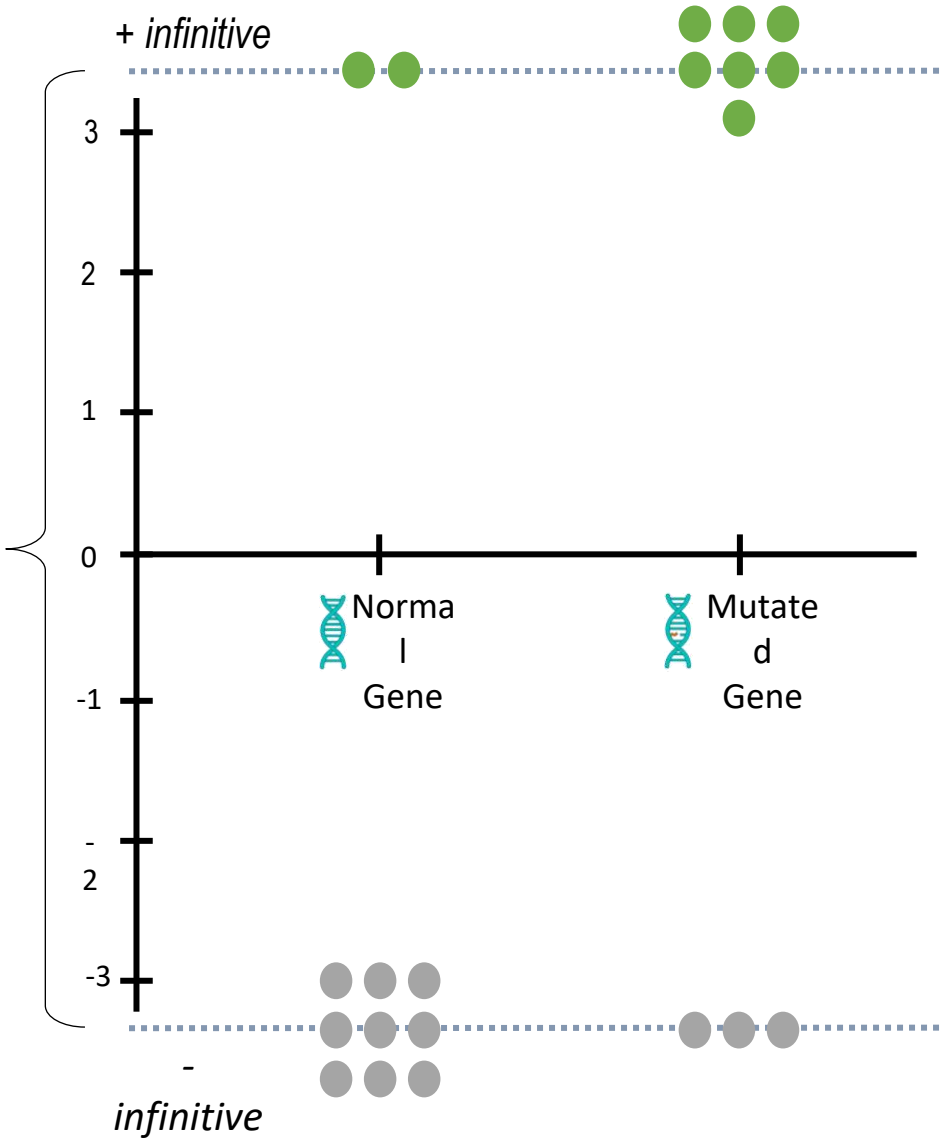
- The coefficient is on the **log-odds scale, not probability,** it can't be read directly as "probability increases by 1.83".
- To interpret it in terms of probability, we need to transform it (next slide).

MICE EXAMPLE – CATEGORICAL VARIABLE

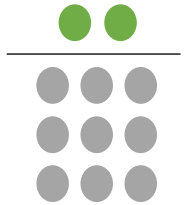
Transform the **y axis** from the **probability** to be obese to the **log(odd of obesity)**



Log (odds of obesity)

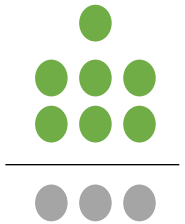


1 Log(odds gene$_{normal}$)



$$\log\left(\frac{2}{9}\right) = -1.5$$

2 Log(odds gene$_{mutated}$)



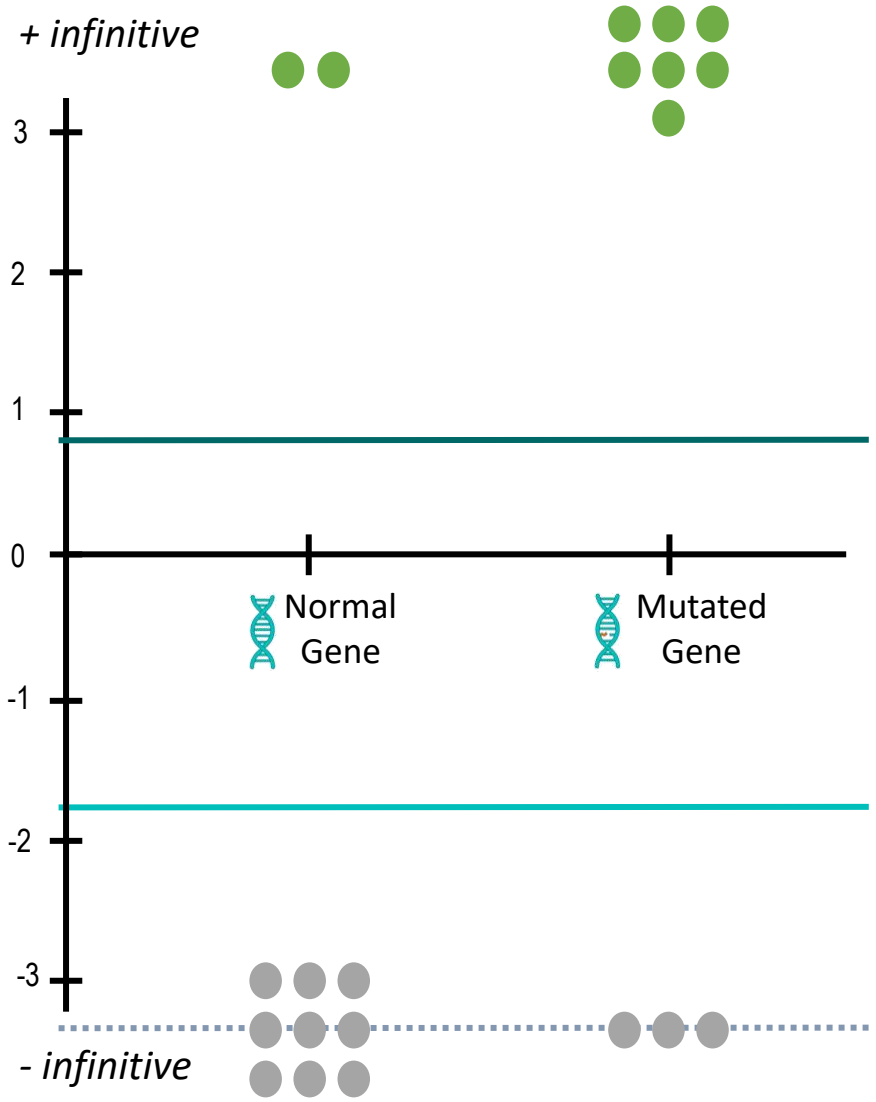
$$\log\left(\frac{7}{3}\right) = 0.85$$

Log (odds of be obese) =
Log(odds gene$_{normal}$) +
(Log(odds gene$_{mutated}$) - Log(odds gene$_{normal}$)) x

Log (odds of be obese) =

Log(odds gene$_{normal}$) + $Log\left(\frac{(\text{odds gene}_{mutated})}{(\text{odds gene}_{normal})}\right) x$

on a log(odds) scale, **how much** having a mutated gene **increases or decreases the odds of being obese**



Example: normal gene vs. mutated gene

	Estimate	Std. Error	z value	Pr(>|z|)
(Intercept)	-1.5041	0.7817	-1.924	0.0544
geneMutated	2.3514	1.0427	2.255	0.0241

$$\hat{y} = -1.50 + 2.35 \cdot \text{geneMutated}$$

- **Reference category**: normal gene (geneMutated = 0). The intercept (−1.50) is the log-odds of obesity for that category.
- The coefficient (2.35) is **how much the log-odds changes when moving from the normal gene to the mutated gene,** not a "per unit" change, but a jump between two groups.
- Every categorical variable in the model needs an **explicit reference category**.

IN OUR DATASET FOR LINEAR REGRESSION

- credito_bancario.csv already has documented reference categories: Asalariado, Casado, Bueno.
- The same principle will apply when we model, for example, tipo_empleo or estado_civil as binary/categorical predictors.

INTERPRETATION

Exponentiating β: from log-odds to Odds Ratio

CONTINUOUS VARIABLE

$$e^{\beta} = e\text{^}(1.83) = 6.2$$

- For each additional unit of weight, the odds of obesity are multiplied by 6.2.
- β > 0 → e^β > 1 → odds increase
- β < 0 → e^β < 1 → odds decrease

CATEGORICAL VARIABLE

$$e^{\beta} = e\text{^}(2.35) = 10.5$$

- The odds of obesity are 10.5 times higher with the mutated gene than with the normal gene (reference).
- The value 1.0 is the neutral point: e^β = 1 means "no effect" relative to the reference.

In both cases, e^β is called the odds ratio (OR) — the standard way to report a predictor's effect in logistic regression.

ESTIMATION

How are the β's estimated? Maximum likelihood

- Unlike OLS, there's no closed-form formula for β here. Instead, we look for the β values that make the observed data most "believable" (likely).

LIKELIHOOD

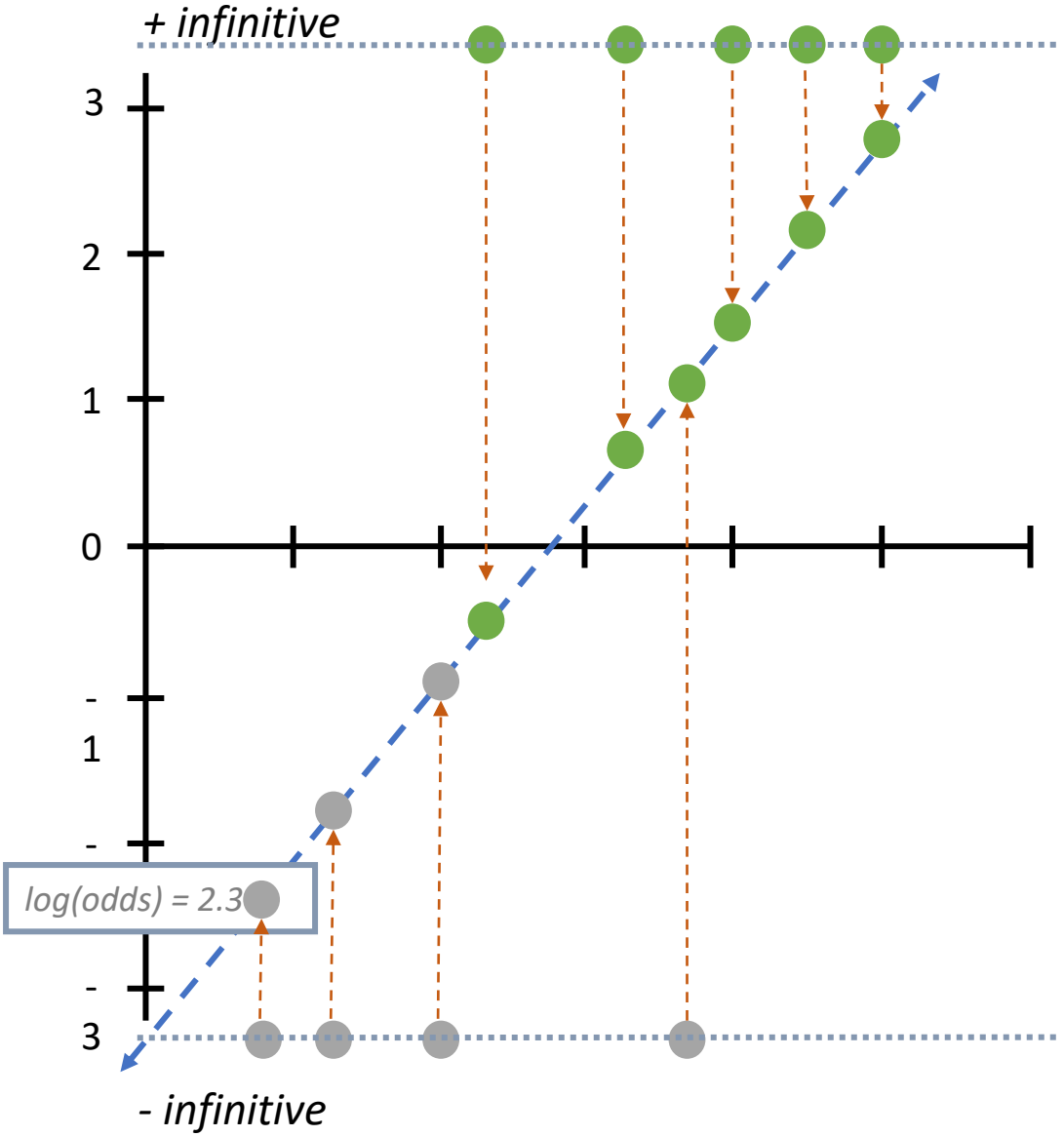
$$L(\beta) = \Pi\ p_i\ ^{(y_i)}\ \cdot\ (1-p_i)\ ^{(1-y_i)}$$

- Each observation contributes p_i if $Y_i=1$, or $(1-p_i)$ if $Y_i=0$.
- The product of all the probabilities is the likelihood of the data, given a candidate β.

LOG-LIKELIHOOD

$$\ln L(\beta) = \Sigma\ [y_i \ln p_i + (1-y_i) \ln(1-p_i)]$$

- Working with logarithms turns the product into a sum (more numerically stable).
- Iterative optimization (Newton-Raphson / IRLS) is used to find the β that maximizes this expression — statsmodels solves it automatically.



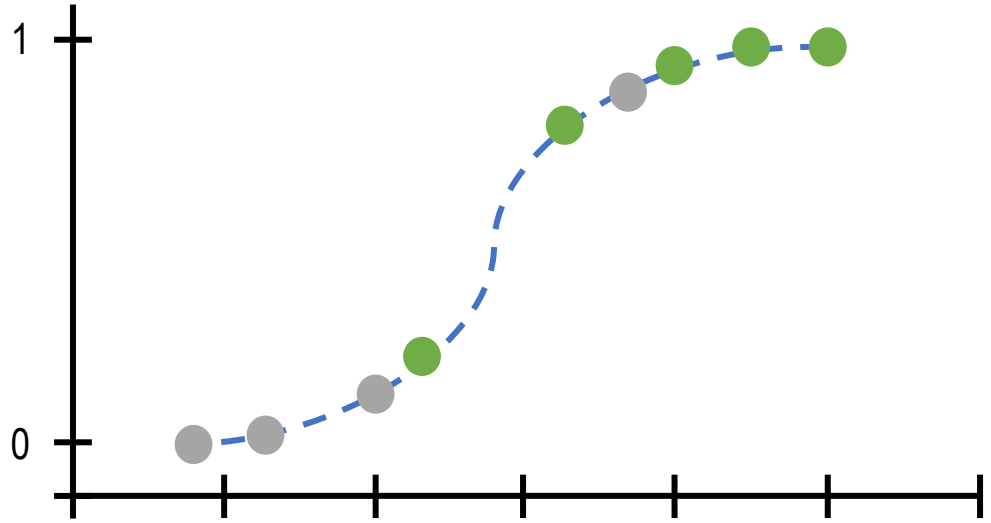
Candidate line

Project the data points in the line

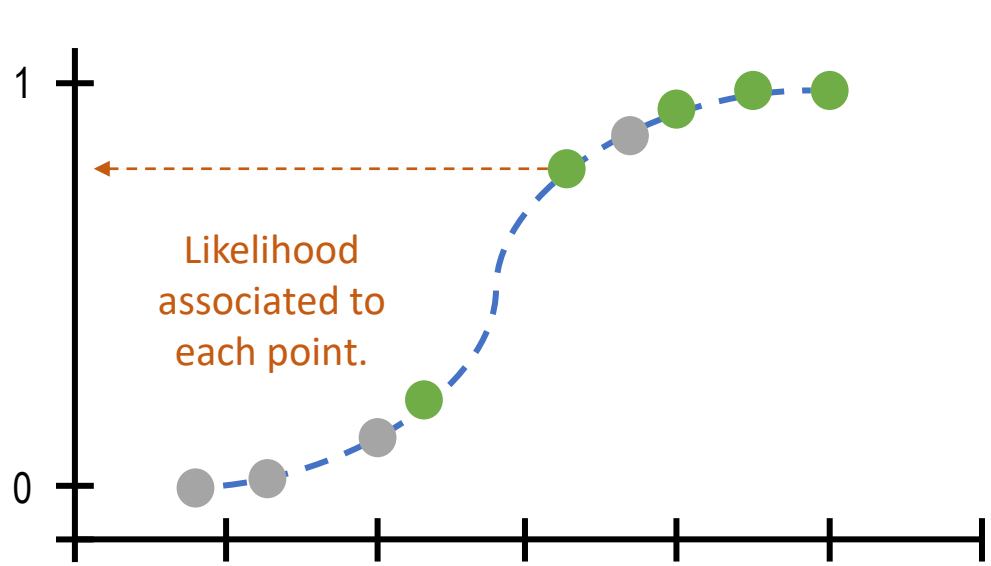
Candidate of log (odds) values

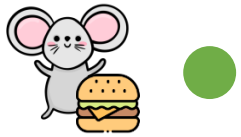
Transform the candidate of log (odds) values to **candidate probabilities.**

$$p = \frac{e^{\log(odds)}}{1 - e^{\log(odds)}}$$



Calculate maximum likelihood







Likelihood of being obese

0.49 x **0.9** x **0.91** x **0.91** x **0.92**

The likelihood of all the obese mise.

Likelihood of not being obese

(1 - 0.9) x **(1 - 0.3)** x **(1 - 0.01)** x **(1 - 0.001)**

The likelihood of all the non obese mise.

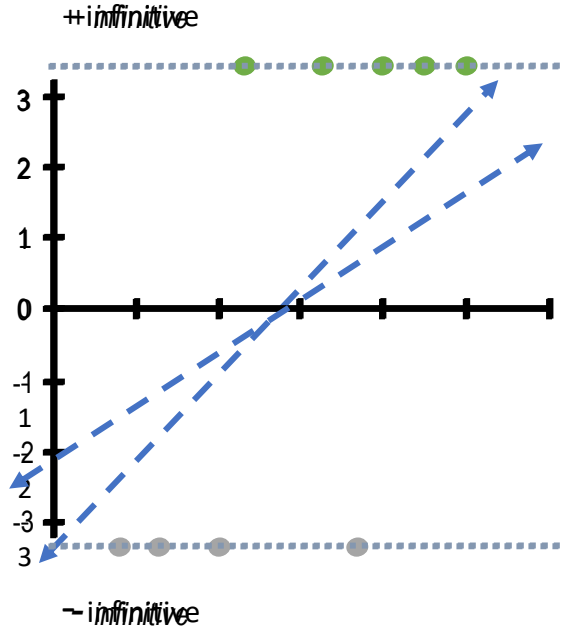
Likelihood of the data given the squiggle =

0.49 x 0.9 x 0.91 x 0.91 x 0.92 x
(1 - 0.9) x (1 - 0.3) x (1 - 0.01) x (1 - 0.001)

Log Likelihood of the data given the squiggle =

log(0.49) + log(0.9) + log(0.91) + log(0.91) + log(0.92) +
log(1 - 0.9) + log(1 - 0.3) + log(1 - 0.01) + log(1 - 0.001)

-3.77



Repeat

INFERENCE

Is the coefficient significant? The Wald test

HYPOTHESES

$H_0: \beta_1 = 0$ (X has no effect)

$H_1: \beta_1 \neq 0$ (X does have an effect)

TEST STATISTIC

$$z = \hat{\beta} / SE(\hat{\beta})$$

Follows a standard normal distribution under H_0

DECISION RULE

- If Pr(>|z|) < 0.05 → H_0 is rejected: the coefficient is statistically significant at the 5% level.
- In the weight example: p = 0.0934 → H_0 is not rejected at 5% (though it would be significant at 10%).
- In the geneMutated example: p = 0.0241 → H_0 is rejected: the mutated gene has a significant effect.

McFadden's Pseudo-R^2

- In linear regression we use R^2 comparing SS(mean) vs. SS(fit). In logistic regression, the equivalent compares log-likelihoods.

LINEAR REGRESSION

$$R^2 = (SS(mean) - SS(fit)) / SS(mean)$$

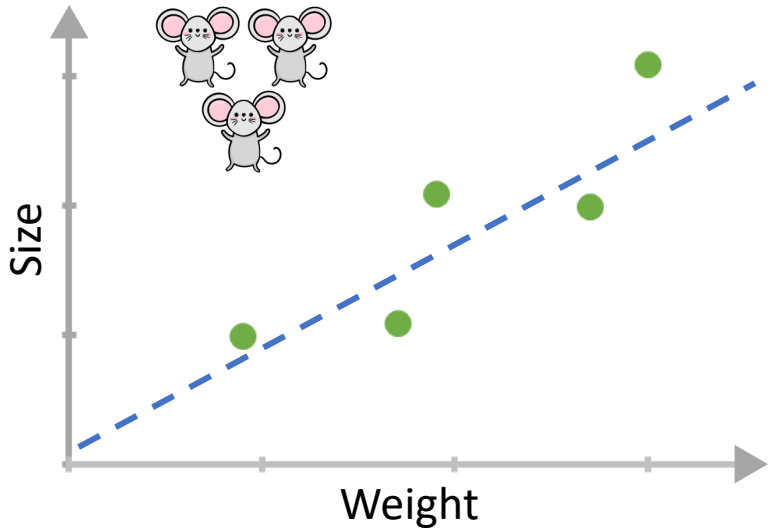
LOGISTIC REGRESSION

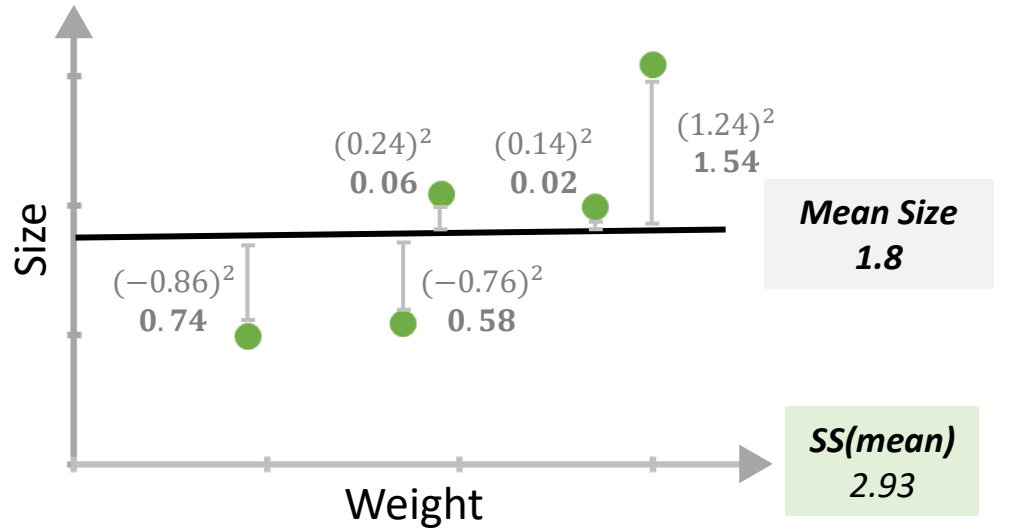
$$R^2 = (LL(base) - LL(fit)) / LL(base)$$

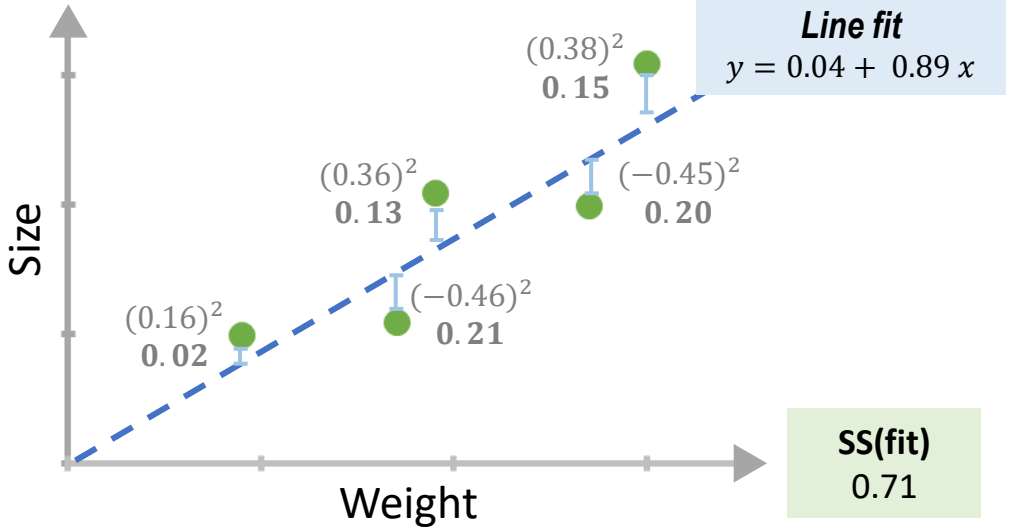
$$R^2 = (-6.18 - (-3.77)) / -6.18 = 0.39$$

- Interpretation: this is NOT "proportion of variance explained" as in OLS, but a relative improvement in likelihood.
- Values between 0.2 and 0.4 are already considered a good fit according to McFadden.

R^2 Coefficient of determination







Does the line fit better tan the mean?
How much better?

$$R^2 = \frac{SS(mean) - SS(fit)}{SS(mean)}$$

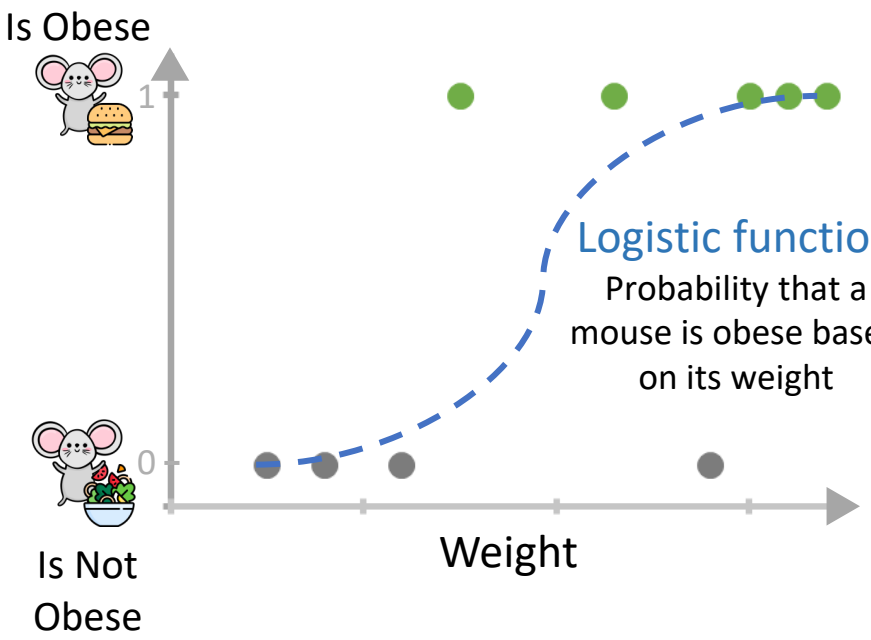
$$R^2 = \frac{2.93 - 0.71}{2.93}$$

$$R^2 = 0.76$$

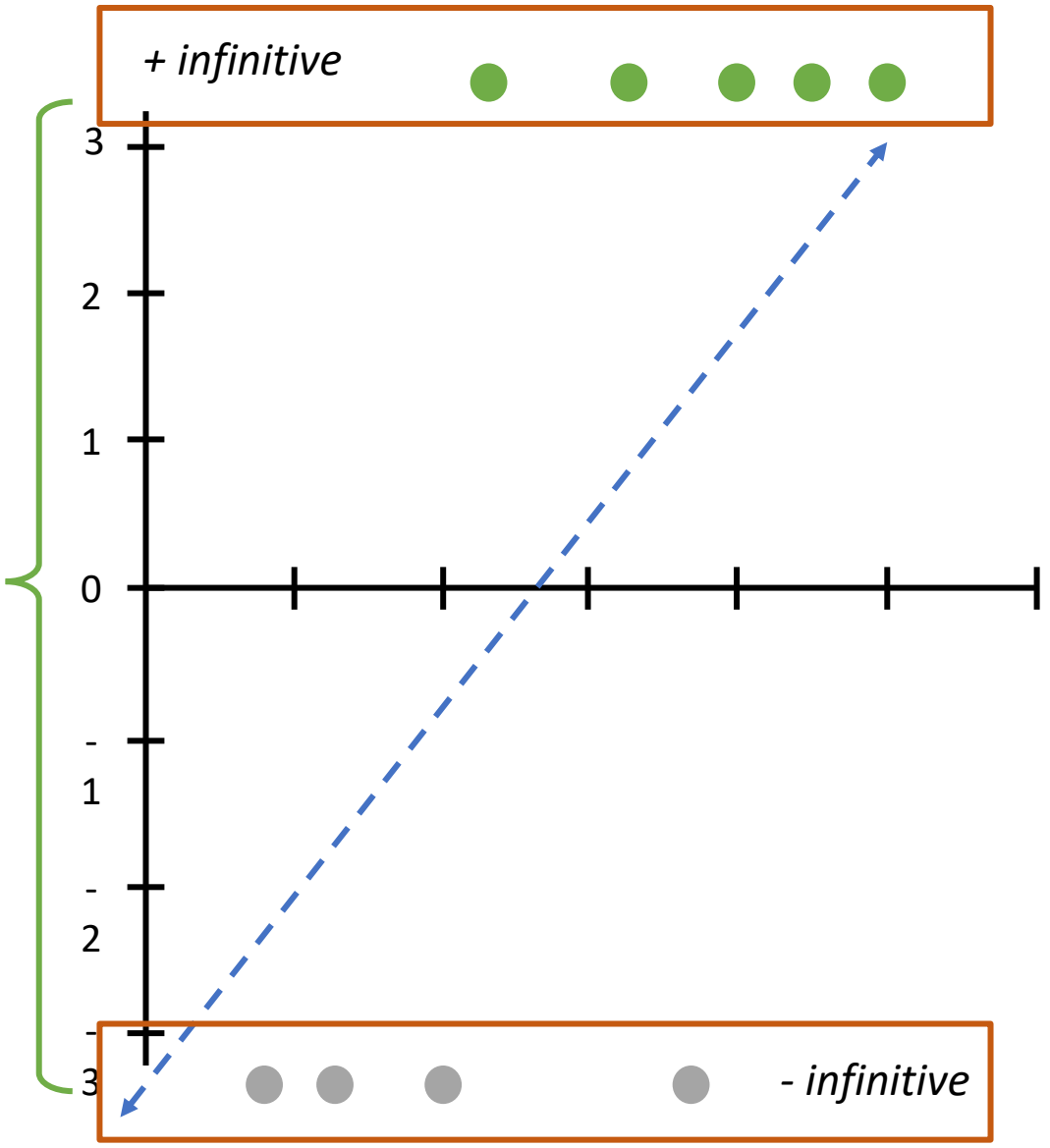
0.76 of the **variation** of the **mouse size** is explain by the **weight**.

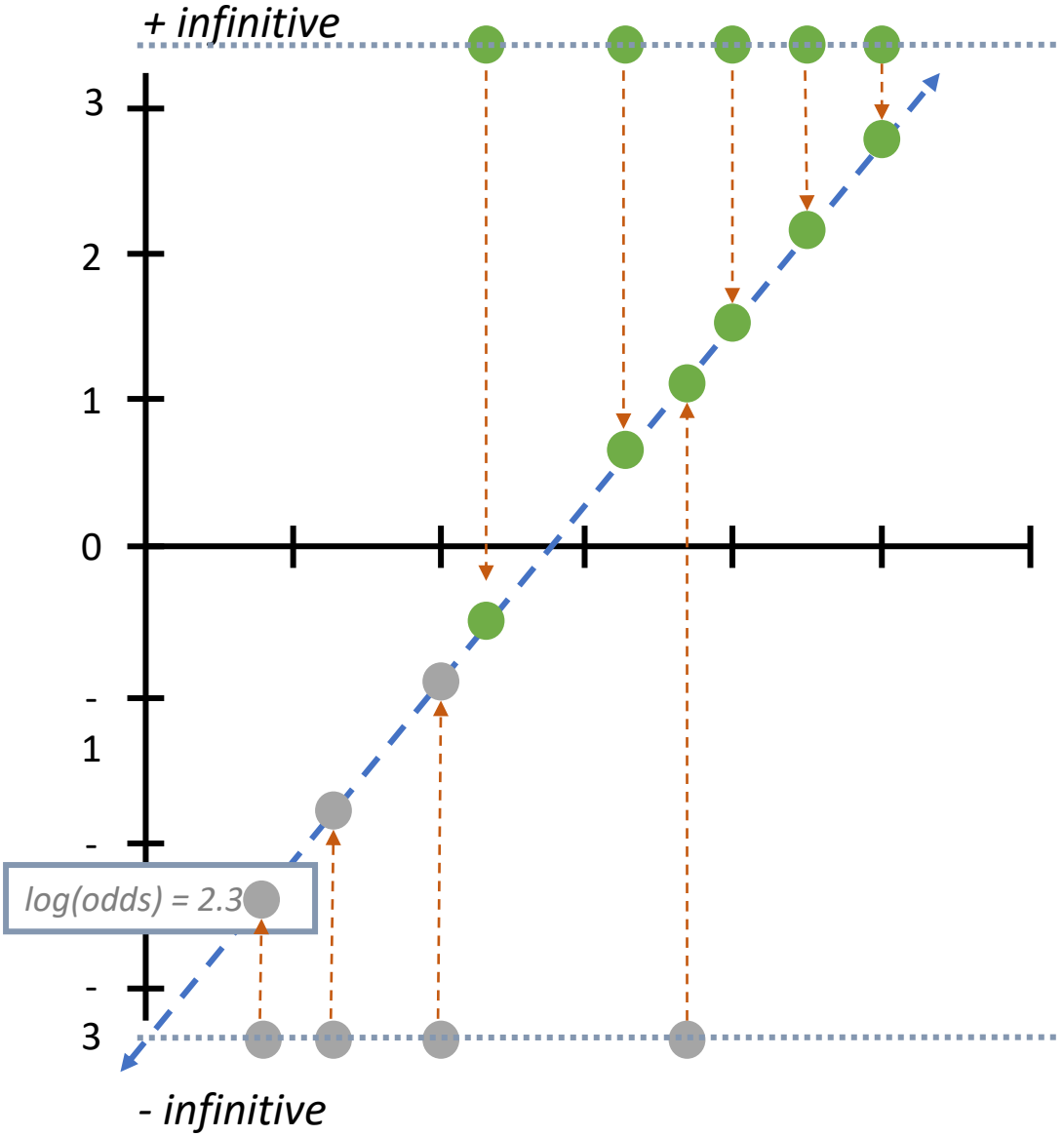
R^2 Coefficient of determination

$$R^2 = \frac{SS(mean) - SS(fit)}{SS(mean)}$$



Log (Odds)





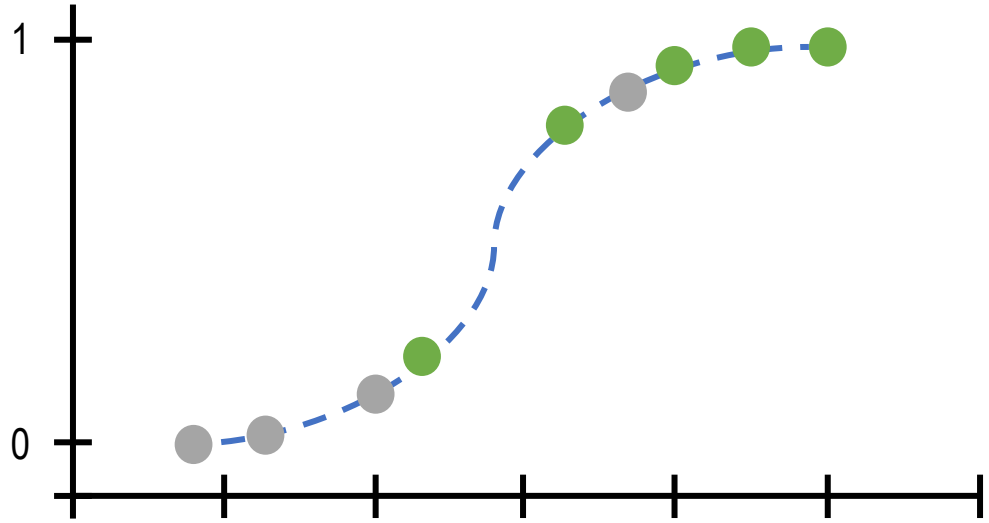
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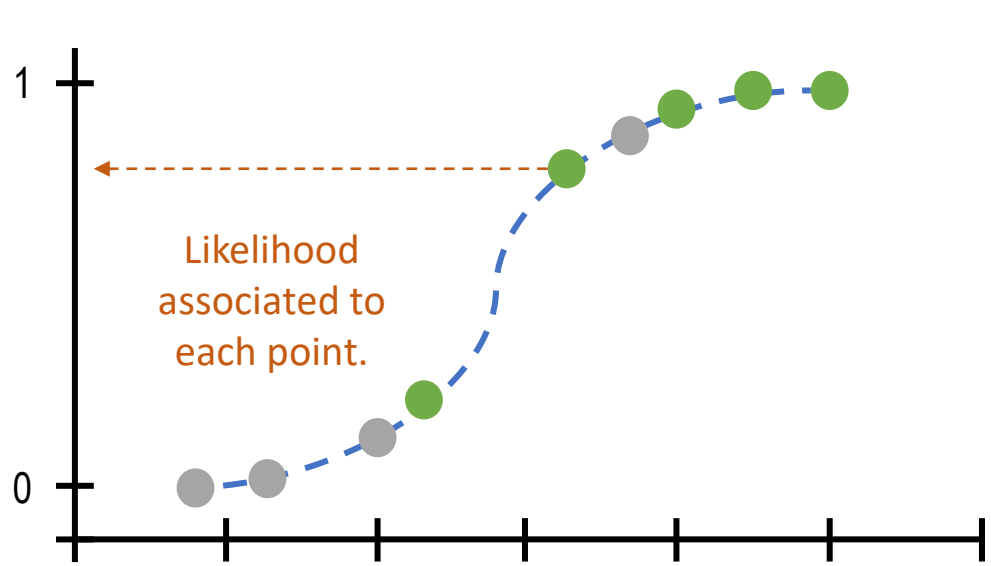
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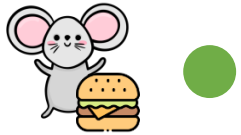
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$$p = \frac{e^{\log(odds)}}{1 - e^{\log(odds)}}$$



Calculate maximum likelihood







Likelihood of being obese

0.49 x **0.9** x **0.91** x **0.91** x **0.92**

The likelihood of all the obese mise.

Likelihood of not being obese

(1 - 0.9) x **(1 - 0.3)** x **(1 - 0.01)** x **(1 - 0.001)**

The likelihood of all the non obese mise.

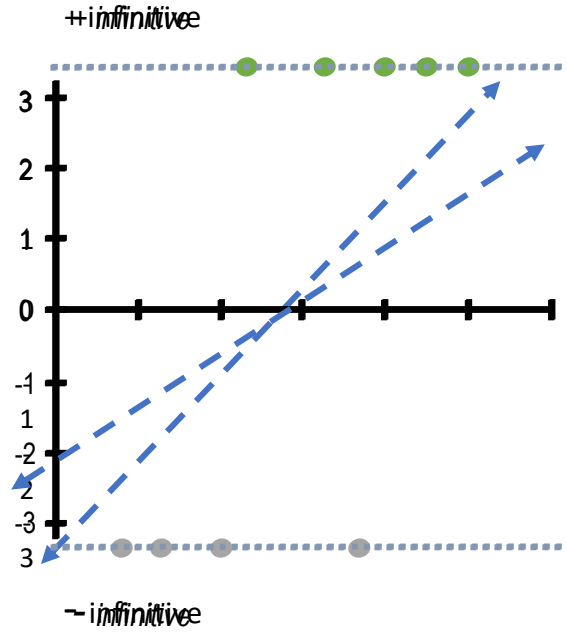
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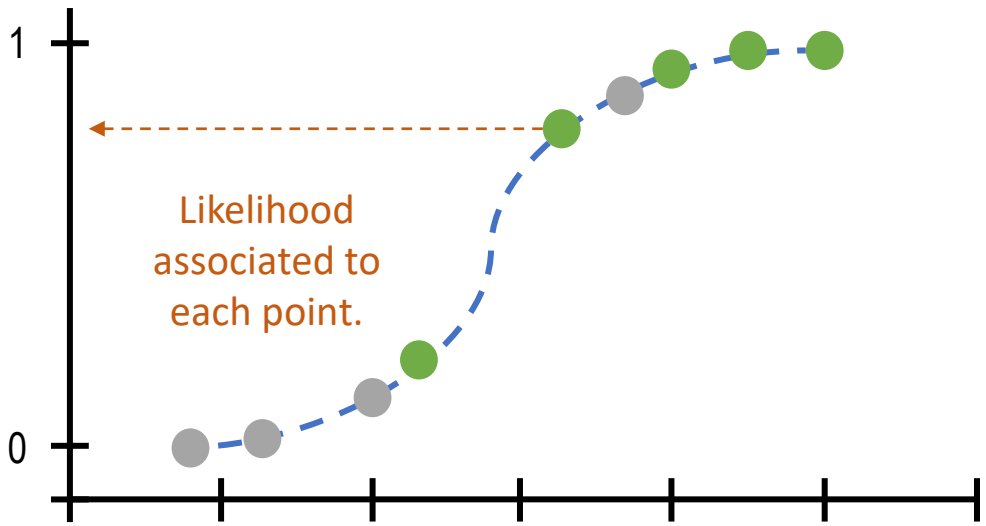
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log(1 - 0.9) + log(1 - 0.3) + log(1 - 0.01) + log(1 - 0.001)

-3.77



Repeat

Logistical function that fits better





Likelihood of being obese

0.49 x **0.9** x **0.91** x **0.91** x **0.92**

The likelihood of all the obese mise.



Likelihood of not being obese

(1 - 0.9) x **(1 - 0.3)** x **(1 - 0.01)** x **(1 - 0.001)**

The likelihood of all the non obese mise.

Log Likelihood of the data given the squiggle =

log(0.49) + log(0.9) + log(0.91) + log(0.91) + log(0.92) +
log(1 - 0.9) + log(1 - 0.3) + log(1 - 0.01) + log(1 - 0.001)

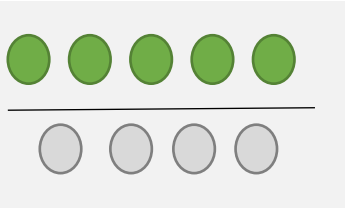
-3.77

$$R^2 = \frac{SS(mean) - SS(fit)}{SS(mean)}$$

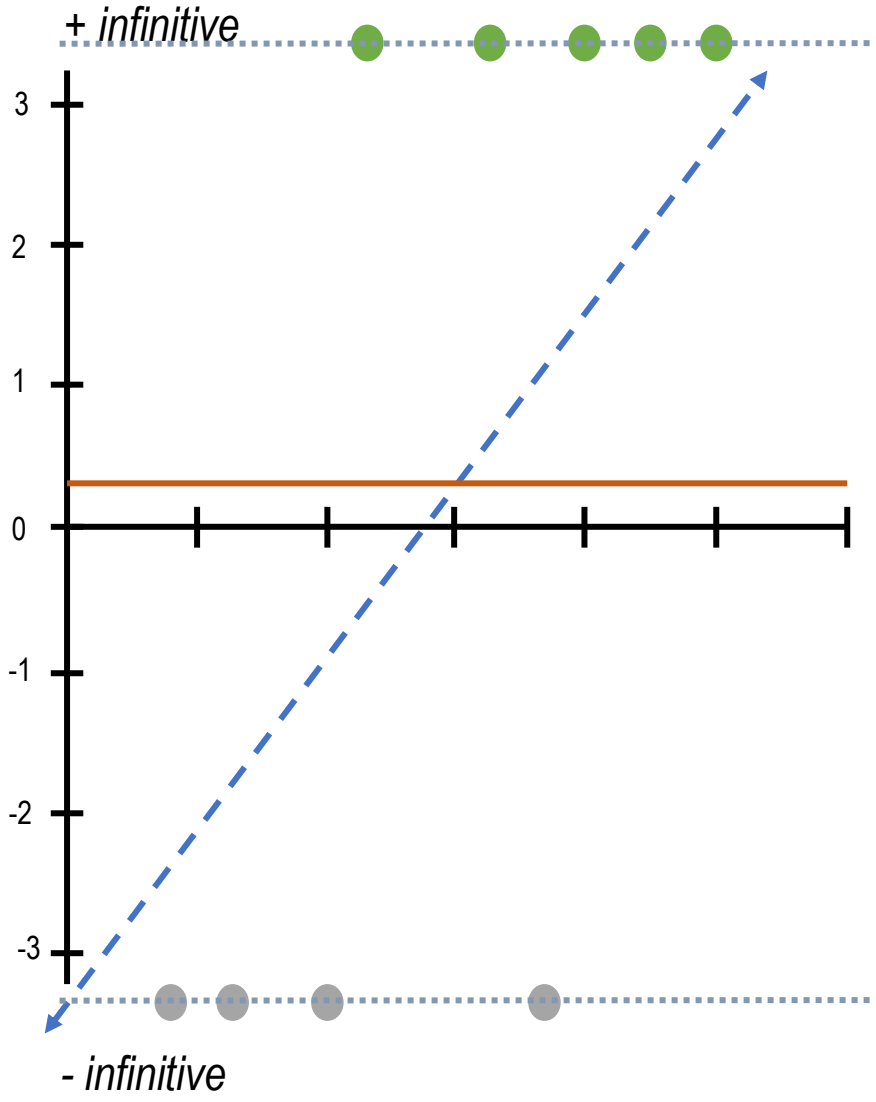
$$R^2 = \frac{? - LL(fit)}{?}$$

LL(mean) analogous to SS(mean)

Lod (odds obesity)
without consider weight



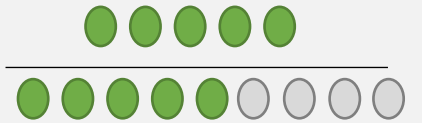
$$\frac{5}{4} = \log\left(\frac{5}{4}\right) = 0.22$$



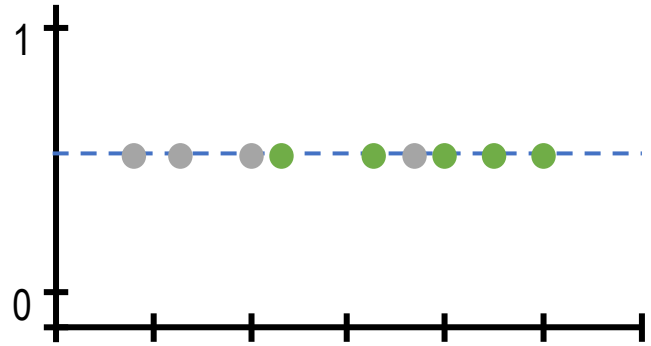
$$p = \frac{e^{\log(odds)}}{1 - e^{\log(odds)}}$$

$$p = \frac{e^{\log(0.22)}}{1 - e^{log(0.22)}} = \mathbf{0.56}$$

Probability of being obese



$$\frac{5}{9} = \mathbf{0.56}$$



Log Likelihood of data given overall probability of obesity =

log(0.55) + log(0.55) + log(0.55) + log(0.55) + log(0.55) + log(1 – 0.55) + log(1 – 0.55) + log(1 – 0.55) + log(1 – 0.55)

-6.18

$$R^2 = \frac{SS(mean) - SS(fit)}{SS(mean)}$$

$$R^2 = \frac{LL(overalll\ prob.) - LL(fit)}{LL(overall\ prob.)}$$

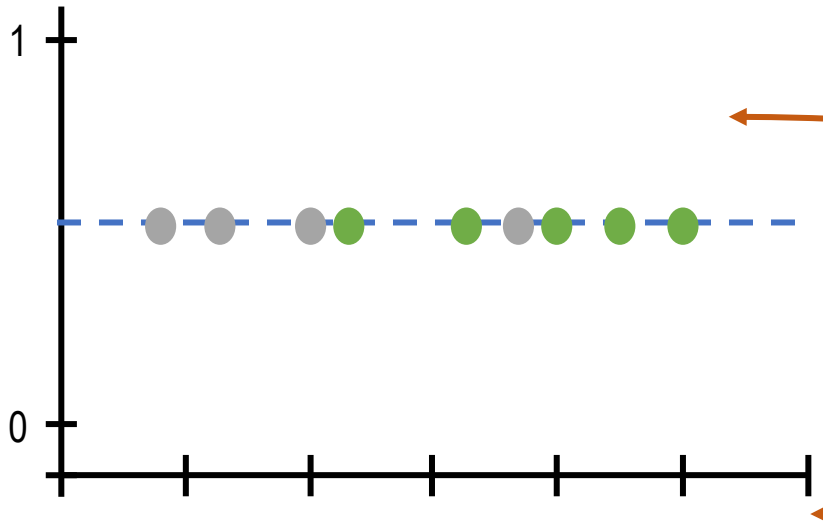
McFadden's Pseudo–R^2

Linear Regression

$$R^2 = \frac{SS(mean) - SS(fit)}{SS(mean)}$$

Likelihood overall probability

Measure of a bad fit

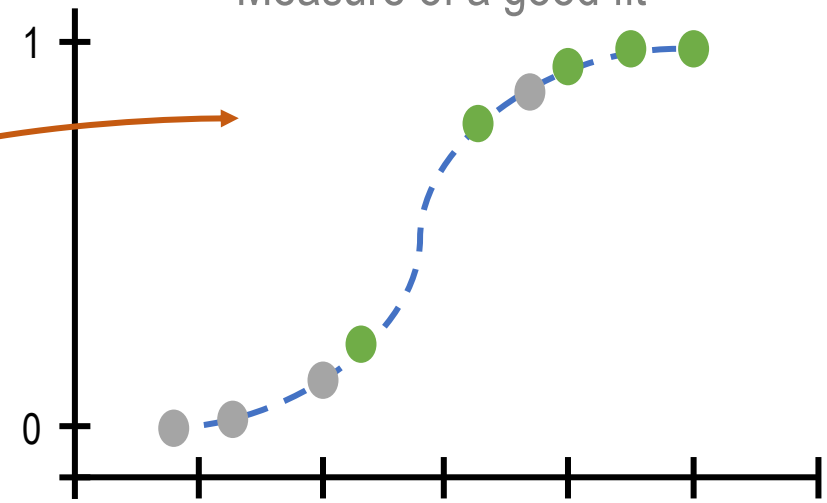


Logistic Regression

$$R^2 = \frac{LL(overall\ prob.) - LL(fit)}{LL(overall\ prob.)}$$

Likelihood Fit

Measure of a good fit



$$R^2 = \frac{-6.18 - (-3.77)}{-6.18} = \mathbf{0.39}$$

EVALUATION

Confusion matrix and classification metrics

	Predicted: Positive	Predicted: Negative
Actual: Positive	**TP**	FN (Type II Error)
Actual: Negative	FP (Type I Error)	**TN**

- **Sensitivity (recall) =** TP / (TP + FN), of those that ARE positive, what % do we detect?
- **Specificity =** TN / (TN + FP), of those that ARE negative, what % do we correctly identify?
- **Precision =** TP / (TP + FP), of those we predicted positive, what % actually were?
- **Negative predicted value =** TN / (TN + FN), of those we predicted negative, what % actually were?
- **Accuracy =** (TP + TN) / total, the overall % of correct predictions.

WHY ISN'T ACCURACY ENOUGH?

- With imbalanced classes (e.g., few delinquent clients), a model can have high accuracy just by always predicting "not delinquent".
- That's why we always report sensitivity and specificity alongside accuracy, especially in credit risk, where the cost of a false negative (an undetected risky client) differs from that of a false positive.